AD 502 – Machine Learning

Pre-Reading Material – Lecture 35

Random Forest-III

Evaluation, Advantages and Applications of Random Forest

Introduction

Random Forest is widely used because it can model complex relationships, handle different types of prediction tasks and provide useful information about feature importance. Its performance must still be evaluated using suitable metrics and unseen data.

Evaluation of Random Forest

The evaluation method depends on the type of problem.

For Classification:

- **Accuracy:** Measures the proportion of correctly classified observations.
- **Precision:** Measures how many observations predicted as positive are actually positive.
- **Recall:** Measures how many actual positive observations are correctly identified.
- **F1-score:** Combines precision and recall into a single metric.

For Regression:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted values.
- **Mean Squared Error (MSE):** Measures the average squared prediction error.
- **Root Mean Squared Error (RMSE):** Represents the square root of MSE and expresses error in the target variable's units.

Feature Importance

Random Forest can estimate the importance of input features in making predictions. Feature importance helps identify variables that contribute strongly to the model's predictive performance.

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For example, in electricity-demand forecasting, historical demand and temperature may be influential features.

However, feature importance does not necessarily establish a causal relationship, and some importance measures can be biased. Permutation importance is an alternative that evaluates the effect of disrupting a feature on model performance.

Advantages of Random Forest

- Supports classification and regression.
- Can capture nonlinear relationships.
- Usually provides better stability than an individual decision tree.
- Can estimate feature importance.
- Can handle datasets containing many input features.

Limitations of Random Forest

- Can require considerable memory and computation when many trees are used.
- Is less interpretable than a single decision tree.
- May need parameter tuning for optimal performance.
- Feature importance measures may be misleading when features are correlated.
- Can perform poorly when training data does not represent real-world conditions.

Practical Applications

Random Forest is applied in healthcare for disease classification, banking for fraud detection, agriculture for crop prediction, manufacturing for fault detection, and the energy sector for electricity-demand forecasting.

For example, a Random Forest regression model can estimate future electricity demand using historical demand, temperature, month and day of the week.

Conclusion

Random Forest is an effective ensemble learning algorithm that combines multiple decision trees. Understanding its construction, evaluation metrics, feature importance, advantages and limitations helps in selecting it for suitable machine learning applications.

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